

Precision Measurement Verification

Measure a set of parts, compare repeated readings, and decide whether the measurements are trustworthy.

Intermediate	1 class period	Skill Builder
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Engineering Context

Replacement rover components arrived without complete inspection records. Your team must verify critical dimensions before the parts are released for assembly.

What You Will Practice

- Use rulers and calipers correctly
- Evaluate repeatability and measurement variation
- Record measurements with appropriate precision
- Make an evidence-based accept/recheck decision

What You Need

- Three to five instructor-selected parts
- Metric ruler
- Digital or dial caliper
- Measurement data sheet
- Engineering notebook

Student Instructions

- 1 Identify the dimensions that are most important to fit or assembly.
- 2 Measure each critical dimension three times without looking at previous results.
- 3 Have a partner independently measure the same dimensions.
- 4 Calculate the range for each set of measurements.
- 5 Compare your readings with the partner readings and any reference dimension provided.
- 6 Mark each dimension Accept, Recheck, or Reject and explain why.
- 7 Write one improvement to the measurement procedure.

Engineering Requirements

- Record units and instrument used for every dimension
- Use three repeated measurements per dimension
- Measurements must be readable and traceable to a labeled feature
- Decisions must cite numerical evidence

Constraints & Safety

- Do not force caliper jaws onto a part
- Use only approved parts
- Do not treat instrument display resolution as guaranteed accuracy

What You Submit

- Labeled part sketch
- Measurement table
- Range calculations
- Accept/recheck/reject decisions
- Procedure improvement note

Reflection Questions

- 1 What caused the largest measurement variation?
- 2 Which instrument was most appropriate for each feature?
- 3 How would temperature or surface damage affect results?
- 4 What measurement would you verify first before assembly?

Engineering Tip

Use the same contact points and instrument orientation each time you repeat a measurement.

Common Mistake

Changing where you measure a feature can create apparent variation that is actually a setup difference.

Stretch Goal

Design a simple go/no-go gauge concept for one critical dimension.

Career Connection

Quality engineers and metrology technicians verify incoming components before they enter aircraft or spacecraft assemblies.